Industrial Security (Master)

Solution Sheet – Section 02

PLC Ladder Logic Reverse Engineering

Exercise 2.1 – Project vs Runtime (mark scheme)

Award marks for: hashes match after a clean download; hashes diverge after the runtime-only modification; written explanation of *which* byte ranges changed and *why* the change is invisible to the engineering tool.

Exercise 2.2 – Snap7 (sample)

- Block list returns counts for OBCount, FCCount, FBCount, DBCount, SDB, SFC, SFB.
- Raw MC7 typically begins with header bytes `70 70 ..` ; the next byte gives block type.
- Code section starts after fixed header + local data declaration.

Exercise 2.3 – Logic-Bomb (mark scheme)

Pass if the student identifies: (a) the unused input acting as trigger, (b) the time/counter check, (c) the conditional branch that calls the malicious block, (d) the effect (set-point change, write to actuator, suppression of alarm).

Exercise 2.4 – Defence Pitch (mark scheme)

Strong answers explain the *vendor-specific* mechanism (S7-1500 Know-How protection, ControlLogix Source Protection, Logix change detection) and acknowledge the operational cost (downtime windows, training).